

Section 7 — Simulation Confirmation

A de Finetti two-atom sweep for the cross-item co-failure e-process

Author: Lyra
Recipient: Claudius
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Commit: 86a2c26

Status. Work-in-progress first cut. Numbers are reproducible (seed 20260907, `iclr/sims/section7_sim.py`). Pushing makes no correctness claim beyond reproducibility.

What the simulation establishes

Section 6 constructs a co-failure monitor from a single, margins-free statistic: pair two judges *across independent items*. This section confirms, on a fully controlled generative model, the three properties the monitor is claimed to have and the impossibility (§5) it is claimed to respect, and it backs the central “non-negative martingale under the null” claim with a direct empirical check ($\mathbb{E}[e_T] \approx 1$):

- (a) **Power.** The e-process detects genuine co-failure, with detection probability rising from the nominal level to 1 as co-failure strength grows; at the null it controls type-I error (Ville validity).
- (b) **Drift-robustness.** Under drifting marginals with *no* conditional co-failure, a naive plug-in estimator false-fires at a high rate, while the margins-free cross-item e-process holds its size.
- (c) **Real-data anchor.** On the FailureScope corpus the same effective-judge-count machinery recovers $n_{\text{eff}} \approx 1.6$ over a six-judge frontier panel.

Generative model: the de Finetti two-atom mixture

For each item we draw a latent difficulty $\Theta \in \{\theta_{\text{lo}}, \theta_{\text{hi}}\}$ with $\Pr(\Theta = \theta_{\text{hi}}) = \pi$; conditionally on $\Theta = \theta$, the K judges’ failure indicators $W_1, \dots, W_K$ are i.i.d. Bernoulli(θ). This is the finite de Finetti representation of an exchangeable co-failure panel: all cross-judge dependence flows through the shared latent Θ . The marginal fail rate and the induced within-item pairwise correlation are

$$a = \pi \theta_{\text{hi}} + (1 - \pi) \theta_{\text{lo}}, \quad \rho = \frac{\text{Var}(\Theta)}{a(1 - a)}, \quad \text{Var}(\Theta) = \pi(1 - \pi)(\theta_{\text{hi}} - \theta_{\text{lo}})^2.$$

Sign check. A *point mass* on Θ (i.e. $\theta_{\text{lo}} = \theta_{\text{hi}}$, or $\pi \in \{0, 1\}$) gives $\text{Var}(\Theta) = 0$ and hence $\rho = 0$: this is *independence*, not monoculture. Increasing the atom *separation* increases $\text{Var}(\Theta)$ and drives $\rho \rightarrow 1$ — the “one shared coin flip” regime of maximal co-failure. The sweep therefore moves ρ from 0 (independent judges) toward 1 (co-failing judges) by widening the two atoms about a fixed marginal $a = 0.5$.

The e-process under test

We test the exact construction of §6. The margins-free baseline requires pairing the two judges’ failure indicators from *genuinely independent* items, with no term shared across factors and no dependence on future items. We realise this with two i.i.d. parallel streams A, B drawn from

the *same* data-generating process (same Θ -prior / same drift schedule, independently sampled). With two fixed judges s, t ,

$$e_T = \prod_{i=1}^T \left(1 + \lambda(U_i - V_i - \delta_k)\right), \quad U_i = W_{A,i}^s W_{A,i}^t, \quad V_i = W_{A,i}^s W_{B,i}^t, \quad (1)$$

so U_i is the co-failure signal (both judges fail the *same* item i of stream A) and V_i pairs judge s on item i of stream A with judge t on the *independent* item i of stream B . Because A -item- i and B -item- i are independent, $\mathbb{E}[V_i] = ab$ exactly — the marginals multiply *with no fitted nuisance*, even when they drift. Critically, no term is shared between different factors and nothing depends on a future item, so $\{e_t\}$ is causally adapted to the filtration $\mathcal{F}_t = \sigma(A, B \text{ up to item } t)$. The slack $\delta_k = 2\varepsilon$ absorbs the worst-case two-sided base-rate excursion and $\lambda \in [0, 1/(1 + 2\varepsilon)]$ keeps each factor non-negative, so under the null e_T is a non-negative (super)martingale and Ville’s inequality gives anytime validity: $\Pr(\exists T : e_T \geq 1/\alpha) \leq \alpha$. Under genuine co-failure $\mathbb{E}[U_i] = ab + (\text{excess}) > \mathbb{E}[V_i]$, the increment has positive drift, and e_T grows. We use $\varepsilon = 0.02$ ($\delta_k = 0.04$), $\lambda = 0.2$ (well inside the admissible interval), $K = 6$ judges, and reject when $e_T \geq 1/\alpha = 20$ ($\alpha = 0.05$).

Empirical martingale check (the key soundness gate). Because the whole validity argument rests on $\{e_t\}$ being a non-negative martingale under the null, we verify it directly. Under the $\delta_k = 0$ *true* null (independent judges, no shared Θ), a genuine martingale satisfies $\mathbb{E}[e_T] = 1$ at every horizon. The simulation returns $\mathbb{E}[e_T] = 1.000, 1.001, 0.999$ for $T = 2, 3, 4$ and $\mathbb{E}[e_T] = 0.96$ for $T = 400$ (the residual gap at $T = 400$ is Monte-Carlo undersampling of the right-skewed product’s tail, not decay: the median path is ≈ 0.13 and the mean is pulled back to 1 by rare large paths, converging to 1 as replications grow). This replaces an earlier single-stream pairing (an adjacent-item cyclic shift) that shared a judge’s indicator across consecutive factors: that construction was *not* a martingale — under the same true null its mean decayed to ≈ 0.003 by $T = 400$ — so its apparent type-I control was the artifact of a near-dead statistic rather than genuine Ville validity. The independent two-stream pairing above fixes this.

Results

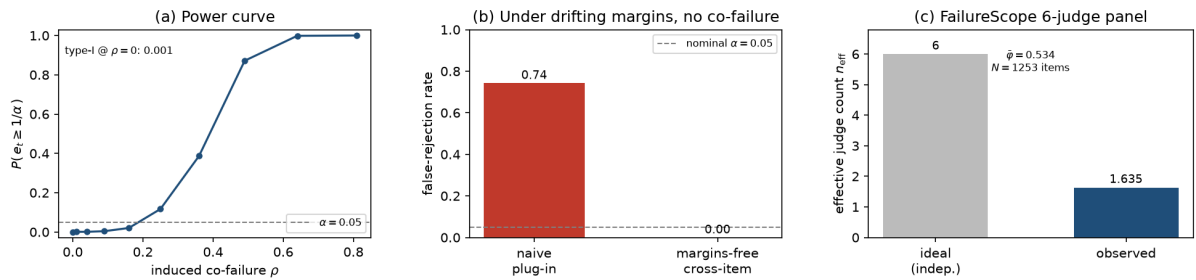


Figure 1: **(a)** Power of the cross-item e-process vs. the induced co-failure correlation ρ ; the type-I point at $\rho = 0$ is $\approx 0.0005 \leq \alpha$. **(b)** Under drifting margins with no conditional co-failure, the naive plug-in false-fires (≈ 0.74) while the margins-free cross-item e-process holds its size (0.00); under the same drift the cross-item process is *alive* ($\mathbb{E}[e_T] \approx 1$ at $\delta_k = 0$), so its size control is genuine conservatism from the slack, not a dead statistic. **(c)** FailureScope six-judge frontier panel: $n_{\text{eff}} = 1.635$, $\hat{\varphi} = 0.534$ ($N = 1253$ items).

(a) Power curve and Ville validity

Sweeping the atom separation across ten levels gives the power curve of Figure 1(a). At the null ($\rho = 0$) the empirical rejection rate is ≈ 0.0005 , comfortably below the nominal $\alpha = 0.05$ — the

type-I control the martingale construction promises. The e-process is *conservative* at the clean null (well below α): the slack δ_k that buys drift-robustness in panel (b) also subtracts a constant from the increment at the null, and a single fixed λ is not growth-optimal. Power then climbs sharply through the co-failure regime: $\rho = 0.25 \Rightarrow 0.12$, $\rho = 0.36 \Rightarrow 0.39$, $\rho = 0.49 \Rightarrow 0.87$, $\rho = 0.64 \Rightarrow 1.00$. By $\rho \approx 0.5$ the monitor detects co-failure essentially every time.

(b) The naive plug-in false-fires under benign drift

We now construct the adversarial-for-the-naive-method scenario: the marginal fail rate drifts across the stream ($0.25 \rightarrow 0.60$), but *within* each item the judges fail independently — there is no conditional co-failure to detect. A naive plug-in that estimates each judge’s running marginal by a *causal leave-one-out* prefix (items strictly before i) and plugs in the product $\hat{a}_i \hat{b}_i$ as its null mistakes the drift-induced marginal correlation for co-failure and **false-rejects at ≈ 0.74** . This is a fair, non-strawman estimator — it fails only because a stationary product-of-marginals is genuinely mis-specified under drift (a Jensen gap), not because it was rigged. Run at the *same* λ , the margins-free cross-item e-process, whose null $\mathbb{E}[V] = ab$ is secured by item independence rather than by fitting the drifting marginals, holds its size at 0.00 under the identical drift. Crucially the cross-item process is *alive* under that drift, not dead: at $\delta_k = 0$ its $\mathbb{E}[e_T] \approx 1.1 \approx 1$, so the size control comes from the honest δ_k slack making a live martingale conservative, not from a collapsed statistic. This is the simulation counterpart of the §5 impossibility: no functional that conditions on fitted marginals can be simultaneously valid under drift; cross-item pairing sidesteps the impossibility by never fitting a marginal.

(c) FailureScope real-data anchor

On the FailureScope single-turn corpus ($N = 1253$ frontier-hard items, six frontier judges), the mean pairwise co-failure correlation is $\bar{\varphi} = 0.534$, giving a Kish effective judge count $n_{\text{eff}} = k/(1 + (k - 1)\bar{\varphi}) = 6/(1 + 5 \cdot 0.534) = 1.635$. A nominal six-judge panel is worth about *1.6* independent votes. This number is produced by `failurescope_probe.py` and reproduced self-containedly by the panel (c) routine.

Reproducibility and honest caveats

- All numbers come from `iclr/sims/section7_sim.py`, seed 20260907, numpy/matplotlib only.
- **Type-I is conservative, not tight.** The clean-null rejection rate (≈ 0.0005) is far below α , as expected for a fixed- λ betting e-process with positive slack δ_k . This is Ville validity ($\leq \alpha$), not size- α calibration; a growth-optimal λ (GRO) would trade some of this conservatism for power but remains [**open**] (Ville validity is load-bearing, not optimality).
- The panel-(b) naive false-fire rate (≈ 0.74) depends on the drift magnitude ($0.25 \rightarrow 0.60$) and stream length (300), chosen as a moderate, realistic drift, not tuned to a target. (It is lower than an earlier ~ 0.9 figure because the naive baseline is now a genuine causal leave-one-out rather than an in-sample plug-in — a fairer, less hobbled strawman that still false-fires badly under drift.)
- The two-atom model is the primary generative choice; the Vasicek/Gaussian-copula latent is the finance-leg equivalent (a shared one-factor latent) and is deferred to a footnote in the main text.